EMATH 100

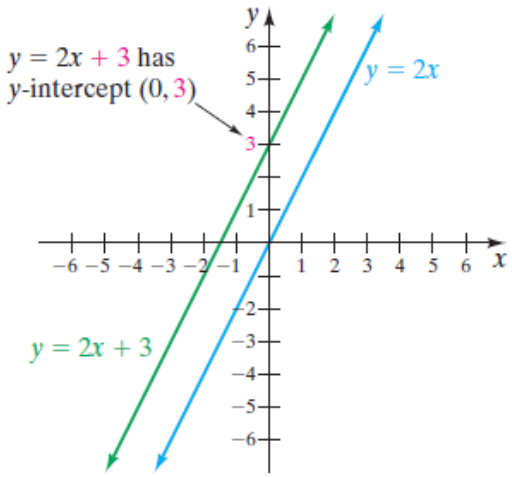
M. PAJA

The Slope-Intercept Form of a Linear Equation

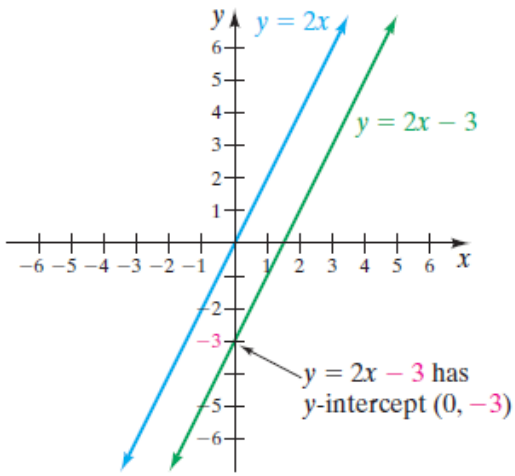
Understand Translations of Graphs

$y = 2x + 3$
$y = 2x$
$y = 2x - 3$

The lines of the graphs of these lines are parallel. We say that the graphs of these equations are vertical translations.



The graph of $y = 2x + 3$ is the graph of $y = 2x$ shifted *up* 3 units.



The graph of $y = 2x - 3$ is the graph of $y = 2x$ shifted *down* 3 units.

Parallel Lines

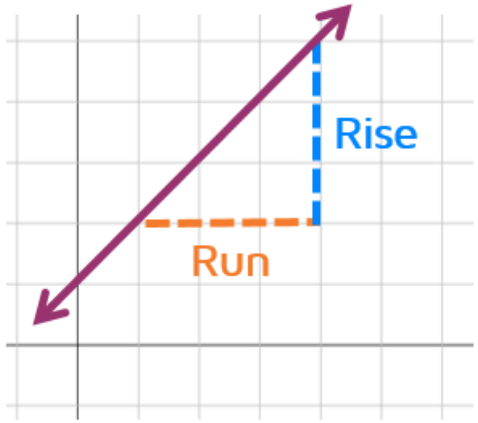
- Parallel lines have the ***SAME slope.***

Find the Slope of a Line

The slope of a line, m, is the ratio of the vertical change, or rise, to the horizontal change, or run, between any two selected points on the line.

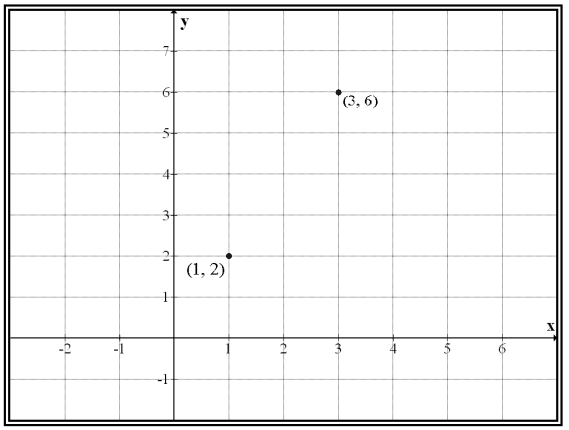
$$\mathbf{m = slope = \frac{vertical\ change}{horizontal\ change} = \frac{rise}{run}}$$

$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{\text{Rise}}{\text{Run}}$$



Find the Slope of a Line

Consider the points
(1,2) and (3, 6)



This means the graph is moving up 4 and to the right 2.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6-2}{3-1} = \frac{4}{2}$$

Find the Slope of the Line

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6-2}{3-1} = \frac{4}{2}$$

Simplifying, $\frac{4}{2} = \frac{2}{1}$, so m = 2

Find the Slope of the Line

Slope of a Line Through the Points (x_1, y_1) and (x_2, y_2)

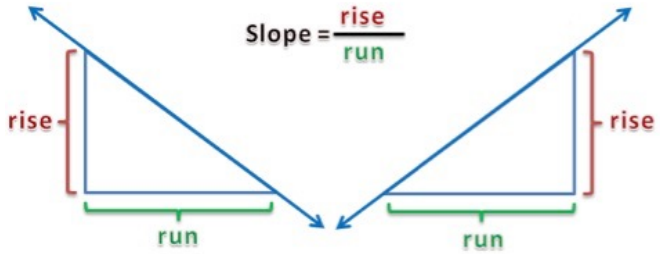
$$\text{m} = \text{slope} = \frac{\text{vertical change}}{\text{horizontal change}} = \frac{rise}{run} = \frac{y_2 - y_1}{x_2 - y_2}$$

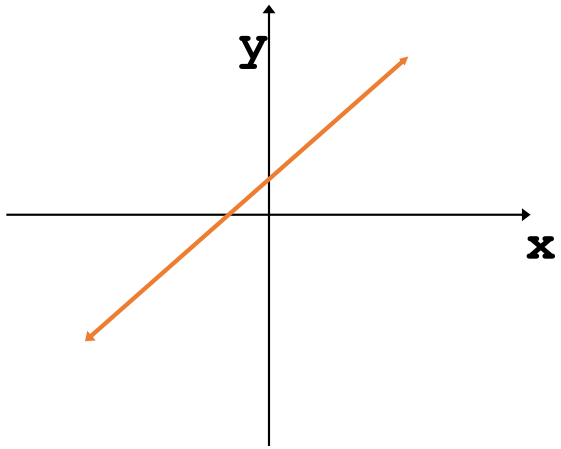
Example

Find the slope of the line with points (-2, 3) and (1, -4).

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-(-4)}{-2-1} = \frac{3+4}{-3} = -\frac{7}{3}$$

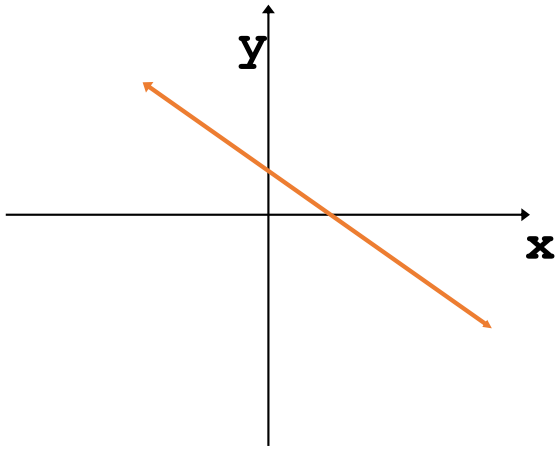
Positive & Negative Slopes





Positive Slope

Line rises from left to right

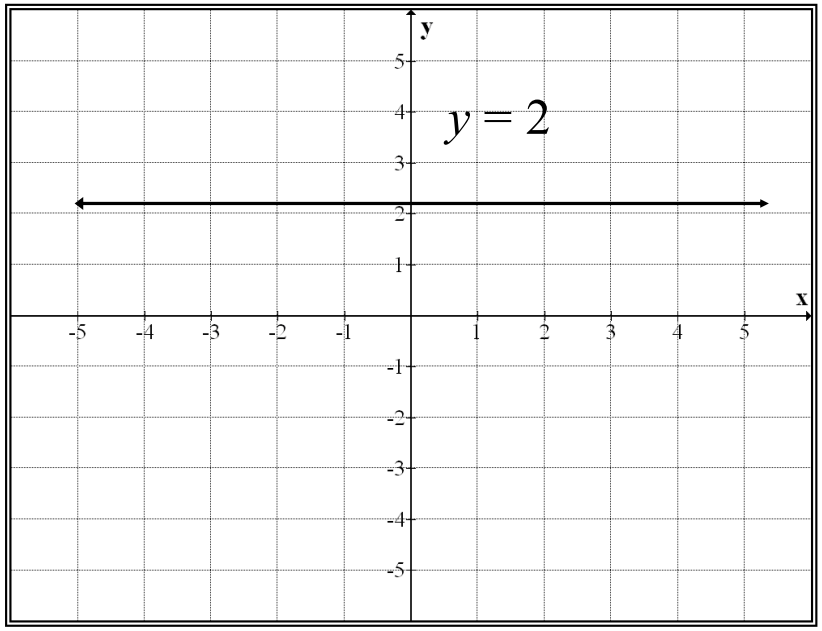


Negative Slope

Line falls from left to right

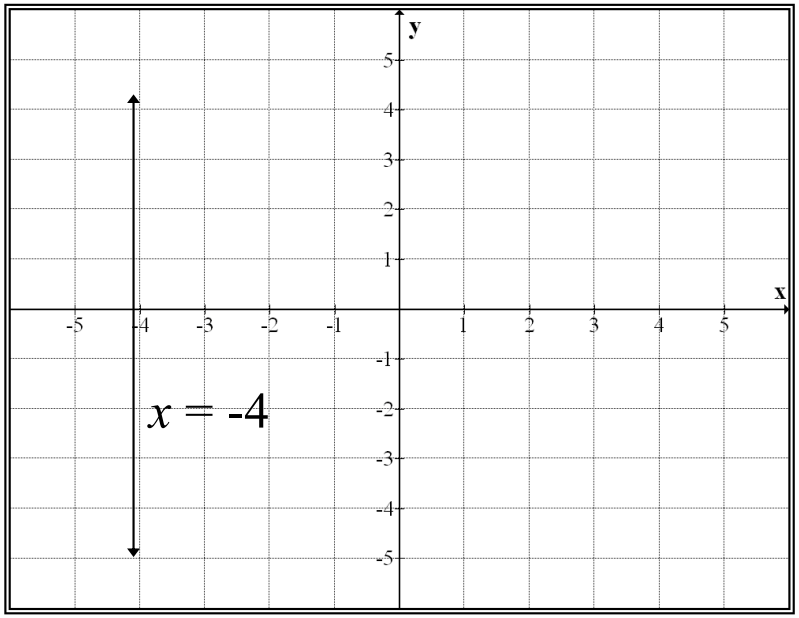
Horizontal Lines

Every horizontal line has a slope of 0.



Vertical Lines

The slope of any vertical line is undefined.



Quick Review

- An equation of a line can be written in slope-intercept form $y = mx + b$

  where

  m is the slope

  b is the y-intercept.

- The y-intercept is where a line crosses the y-axis.

Writing an Equation of a Line

- Suppose the slope of a line is 5 and the y-intercept is 2. How would this you write the equation of this line in slope-intercept form?
- First write the slope-intercept form.

  $y = mx + b$
- Now substitute 5 for m and 2 for b.

  $y = 5x + 2$

Slope-Intercept Form

In the **slope-intercept form**, the graph of a linear equation will always be a ***straight line*** in the form **y = mx + b** where **m** is the slope of the line and **b** is the y-intercept **(0, b)**.

slope y-intercept

$$y = mx + b$$

Examples:

$$y = 3x - 6$$

slope is 3

y-intercept is (0, -6)

$$y = \frac{1}{2}x + \frac{3}{2}$$

slope is 1/2

y-intercept is (0, 3/2)

Slope-Intercept Form

Example **Write the equation -5x + 2y = 8 in slope-intercept form.**

$$-5x+2y=8$$

$$2y=5x+8$$

$$y=\frac{5x+8}{2}$$

$$y=\frac{5x}{2}+\frac{8}{2}$$

$$y=\frac{5}{2}x+4$$

The slope is 5/2; the y-intercept is (0, 4).

If we are given the Slope and a Point

- **Step 1:**
  - **First find the y-intercept. Substitute the slope m and the coordinates of the given point (x, y) into the slope-intercept form, $y = mx + b$. Then solve for the y-intercept b.**

- **Step 2:**
  - **Then write the equation of the line. Substitute the slope m and the y-intercept b into the slope-intercept form, $y = mx + b$.**

If We Are Given the Slope and a Point

- Suppose we have a slope of -3 and it passes through the point (1, 2).
  - We first need to find the y-intercept. We can do this by substituting our information into slope-intercept form and solving for b.

$y = mx + b$

$2 = -3(1) + b$

$2 = -3 + b$ Add 3 to both sides.

$5 = b$ Now we know that the y-intercept is 5.

$y = mx + b$

$y = -3x + 5$

Try This!

- Suppose we have a line with a slope of -1 and passes through the point (3, 4), determine the equation of the line and graph it.

- **Suppose we have a line with a slope of -1 and passes through the point (3, 4), determine the equation of the line and graph.**

$$y = mx + b$$
$$4 = (-1)3 + b$$
$$4 = -3 + b$$
$$7 = b$$
$$y = mx + b$$
$$y = (-1)x + 7$$
$$y = -x + 7$$

PERPENDICULAR EQUATIONS

Parallel Lines

- Parallel lines have the ***SAME slope.***

- **Perpendicular Lines** – lines that intersect at right angles.

- Perpendicular lines *have slopes that are opposite reciprocals of each other.* (Change the sign and flip the fraction)

- Vertical and horizontal lines are perpendicular to each other.

Ex. Identify the slope of the following equation and then identify the parallel and perpendicular slopes.

$$y = \frac{1}{2}x - 7$$

$$m = \frac{1}{2}$$

$$\parallel m = \frac{1}{2} \qquad \perp m = -\frac{2}{1} = -2$$

Ex. Identify the slope of the following equation and then identify the parallel and perpendicular slope.

$$\frac{3y}{3} = \frac{-2x}{3} + \frac{6}{3}$$

$$y = -\frac{2}{3}x + 2$$

$m = -\frac{2}{3}$ $\quad \| \; m = -\frac{2}{3}$ $\quad \perp m = \frac{3}{2}$

Ex. Identify the slope of the following equation and then identify the parallel and perpendicular slope.

$$5x + 4y = 12$$

$$m = -\frac{5}{4} \qquad \| \; m = -\frac{5}{4} \qquad \perp \; m = \frac{4}{5}$$

Ex. Write the slope-intercept form of a line through (7, -1) and parallel to the graph of

$$y = \frac{2}{7}x - 5$$

$$m = \frac{2}{7}$$

$$\| \; m = \frac{2}{7}$$

$$y - y_1 = m(x - x_1)$$

$$y + 1 = \frac{2}{7}(x - 7)$$

$$y + 1 = \frac{2}{7}x - 2$$

$$-1 \qquad -1$$

$$y = \frac{2}{7}x - 3$$

Ex. Write the slope-intercept form of a line through (4, -1) and perpendicular to the graph of

$$y = \frac{2}{7}x - 5$$

$$m = \frac{2}{7}$$

$$\perp m = -\frac{7}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y + 1 = -\frac{7}{2}(x - 4)$$

$$y + 1 = -\frac{7}{2}x + 14$$

$$-1 \qquad -1$$

$$y = -\frac{7}{2}x + 13$$

Ex. Write the slope-intercept form of a line parallel to the graph of 2y + 5x = 2 that passes through (0, 6).

$$2y + 5x = 2$$
$$-5x \quad -5x$$
$$\frac{2y}{2} = \frac{-5x}{2} + \frac{2}{2}$$
$$y = -\frac{5}{2}x + 1$$
$$m = -\frac{5}{2}$$
$$\| \; m = -\frac{5}{2}$$

$$y - y_1 = m(x - x_1)$$
$$y - 6 = -\frac{5}{2}(x - 0)$$
$$y - 6 = -\frac{5}{2}x$$
$$+6 \quad +6$$
$$y = -\frac{5}{2}x + 6$$

Ex. Write the slope-intercept form of a line perpendicular to the graph of 2y + 5x = 2 that passes through (0, 6).

$$2y + 5x = 2$$

$$-5x \quad -5x$$

$$\frac{2y}{2} = \frac{-5x}{2} + \frac{2}{2}$$

$$y = -\frac{5}{2}x + 1$$

$$m = -\frac{5}{2}$$

$$\perp m = \frac{2}{5}$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = \frac{2}{5}(x - 0)$$

$$y - 6 = \frac{2}{5}x$$

$$+6 \quad +6$$

$$y = \frac{2}{5}x + 6$$

$-\frac{5}{2} \cdot \frac{2}{5} = \frac{-10}{10} = -1$

Determine if the following equations are parallel, perpendicular or neither

- Ex 16

$2y + 5x = 2$

$y = \frac{2}{5}x - 4$

$m = -5/2$

$m = 2/5$

$2y + 5x = 2$

$-5x \quad -5x$

$2y = -5x + 2$

$y = \boxed{-5/2}x + 1$

⊥

Ex 17

$y = \frac{2}{7}x - 5$

$3y = -2x + 6$

$m = 2/7$

$m = -2/3$

$\frac{3y}{3} = \frac{-2x}{3} + \frac{6}{3}$

$y = -\frac{2}{3}x + 2$

neither

THE END.